Scalability & Future Plan

Date	11 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-Node Deployment	Render's free tier causes "cold starts" after inactivity and limits the number of concurrent user requests.	High
2	Static Training Data	The model (hdi_model.pkl) is trained on historical 2021 data; predictions may gradually lose real-world accuracy as global economies shift.	Medium
3	Lack of User Accounts	Users cannot save their prediction history, compare past inputs, or log back in to view previous reports.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Hosted on a single free-tier Render instance.	Upgrade to a paid cloud tier (e.g., AWS EC2 or Render Pro) and implement Auto-Scaling to handle traffic spikes.
2	Data Storage	Stateless architecture; no user data or logs are saved persistently.	Integrate a relational database (like PostgreSQL) to store user profiles, prediction histories, and historical UNDP datasets.
3	Performance	Synchronous processing via Flask routing.	Implement Redis and Celery for asynchronous task queuing, especially useful when generating bulk predictions or PDF reports.
4	Security	Basic frontend and backend input validation.	Implement OAuth 2.0 for secure user logins and integrate API rate-limiting (Flask-Limiter) to prevent DDoS attacks or API abuse.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated PDF Report Generation	1-2 Months	Allows policymakers to instantly download, print, and share formal, formatted HDI analysis reports based on their inputs.
Phase 3	Automated Model Retraining (MLOps)	3-4 Months	The backend will automatically fetch new UNDP data via an API and retrain the Linear Regression model, ensuring predictions remain highly accurate over time.
Phase 4	Global Interactive Heatmap Dashboard	6 Months	Upgrades the UI to include a visual, map-based interface allowing researchers to click on countries and compare HDI metrics visually across borders.